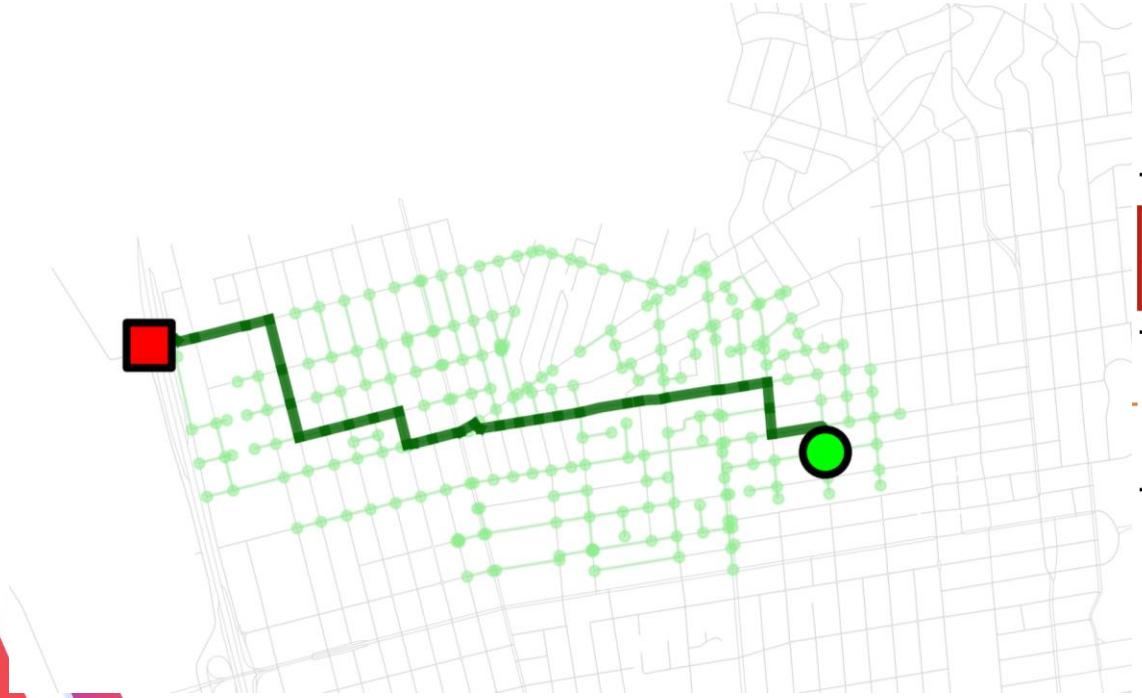
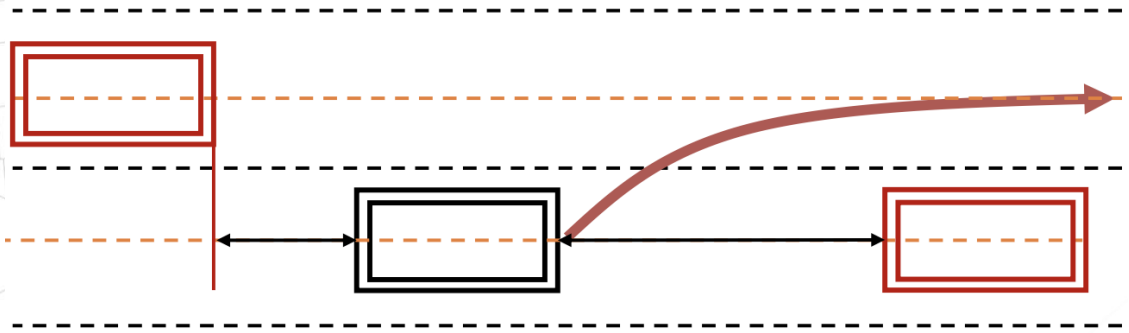
香港城市大學
City University of Hong Kong
Innovating into the Future

CS5493: Tutorial 9
Path Planning

Introduction to Planning

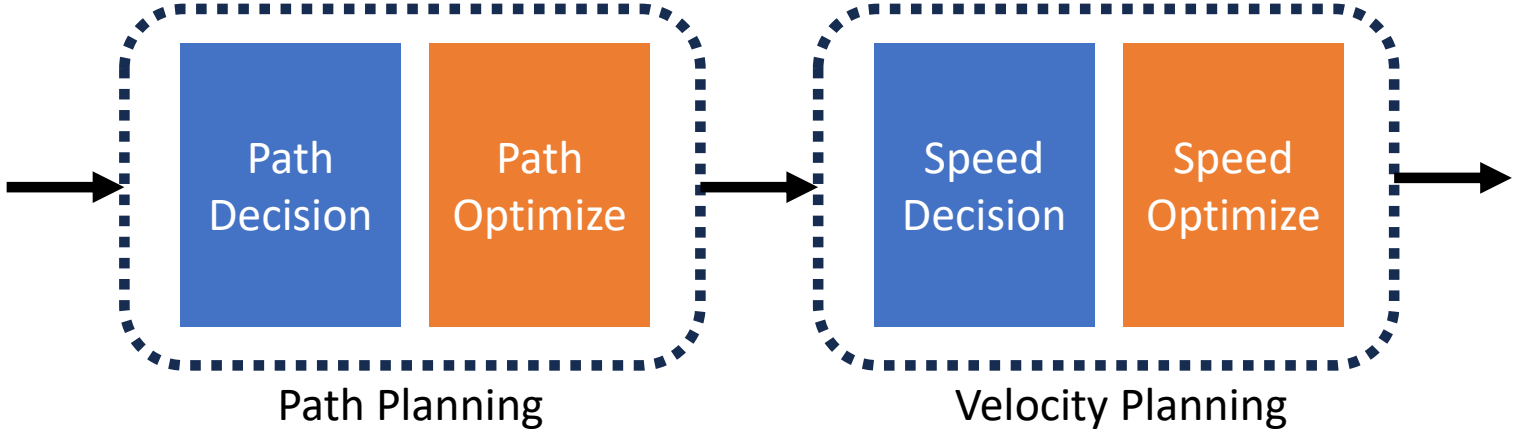
Planning Module focuses on two different scales of path planning.
First is **route navigation**, which generates the route from A to B on a HD Map.
Then, subtle decisions will be made in **trajectory planning**, which focuses on avoiding the obstacles on the planned path, which lane to drive, etc.





Trajectory Planning: Two Steps

In trajectory planning, both spatial and temporal trajectory should be considered.
For spatial trajectory planning we call it path planning, which will handle **static environment** (lanes, obstacles at low speed, …).
For temporal trajectory we define as speed planning which will handle dynamic obstacles.



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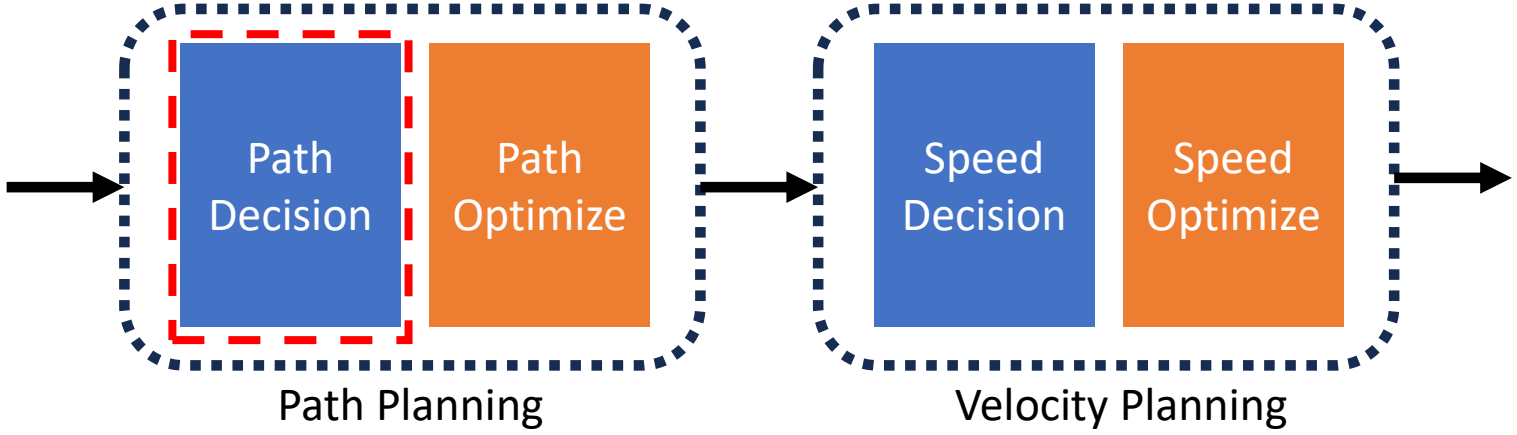
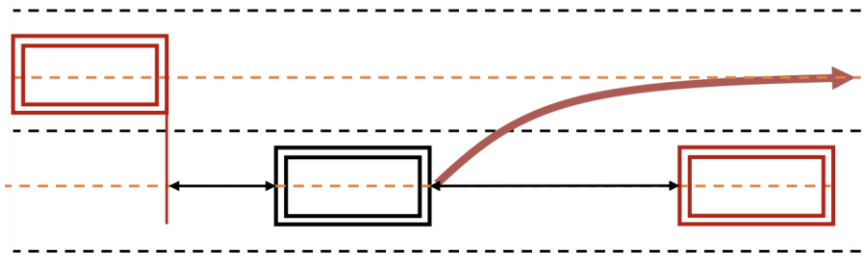


Path Planning and Optimization

LANE_CHANGE_DECIDER: whether change lane



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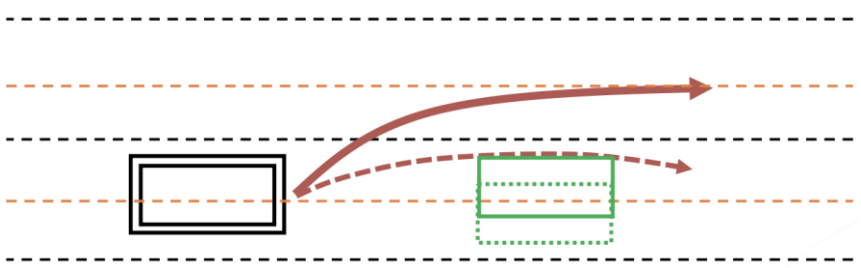
```

Path Planning and Optimization

PATH_REUSE_DECIDER:
whether previous planned lane can be reused



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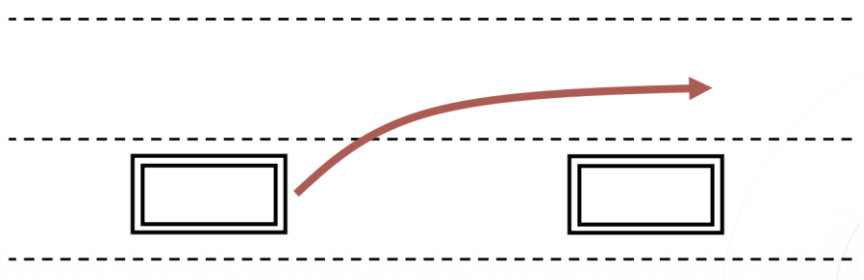
Path Planning and Optimization

PATH_LANE_BORROW_DECIDER:
whether borrow lane?



Check:

1. Whether only one lane per direction
2. Whether there are obstacles ahead
3. Whether the obstacle is far from intersection
4. Whether the obstacle is still
5. Whether it is a straight line or a dashed line between this lane and the opposite lane

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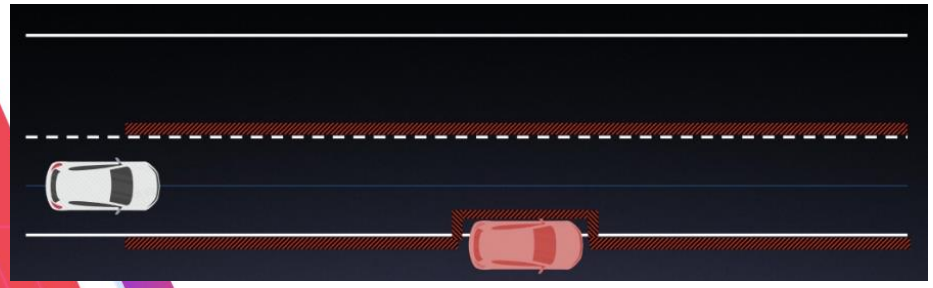
Path Planning and Optimization

PATH_BOUNDS_DECIDER:
generate the boundary of drivable space

Driving on the current lane(/regular/self)

Driving on the current lane with nudge obstacle (/regular/self)



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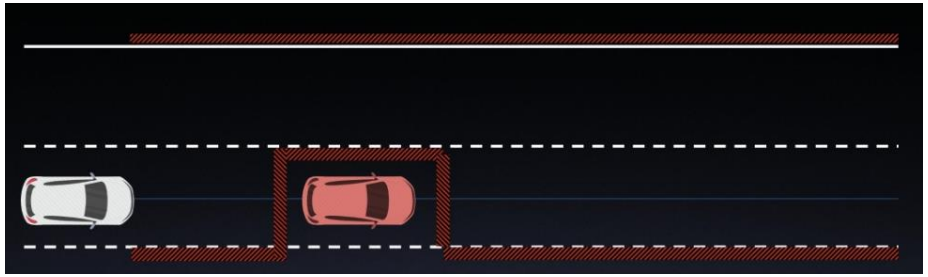
Path Planning and Optimization

PATH_BOUNDS_DECIDER:
generate the boundary of drivable space

Move to the left to overtake and pass.(/regular/left)



Change lane(/regular/lanechange)

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Path Planning and Optimization

PIECEWISE_JERK_PATH_OPTIMIZER:
Based on the drivable boundary, generate optimal path for each boundary using

```
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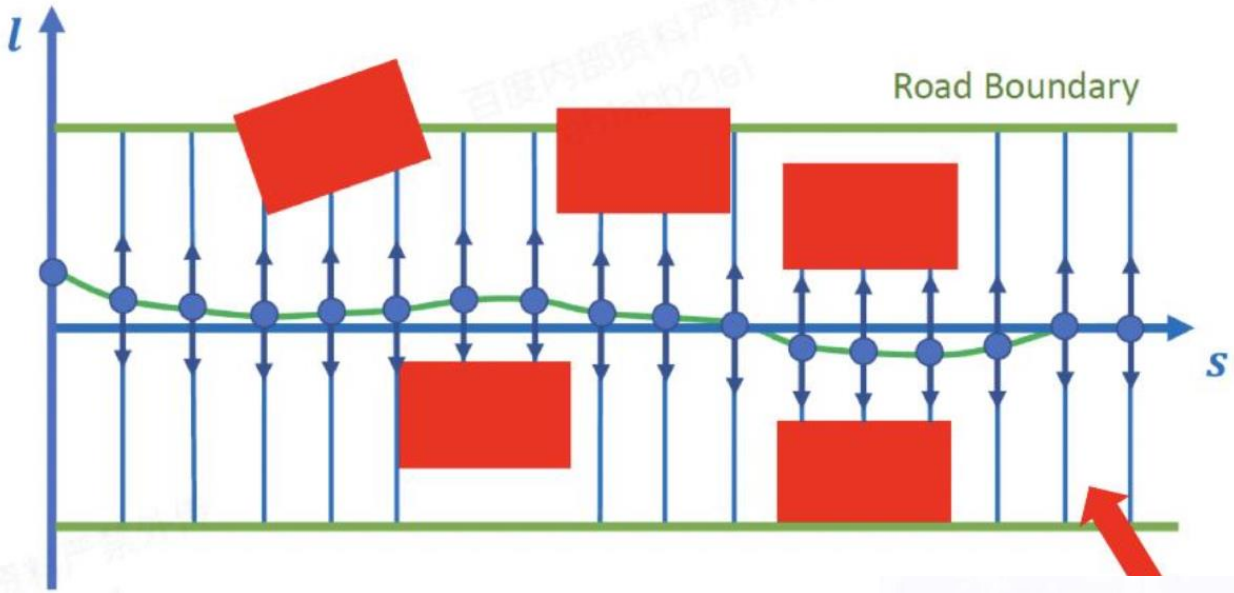
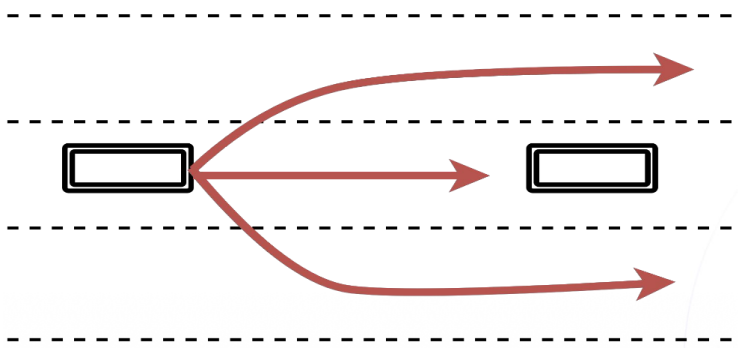


- $f = w_l \sum_{i=0}^{n-1} l_i^2 + w_{l'} \sum_{i=0}^{n-1} {l_i'}^2 + w_{l''} \sum_{i=0}^{n-1} {l_i''}^2 + w_{l'''} \sum_{i=0}^{n-2} {l_{i\to i+1}'''}^2$

Path Planning and Optimization

PATH_ASSESSMENT_DECIDER: assess the quality of different planned lanes.



Whether collision will occur
Length of the planned path
Whether will stop at the overtaking lane
Distance of the path to the ego vehicle
Which lane will return to the current lane first

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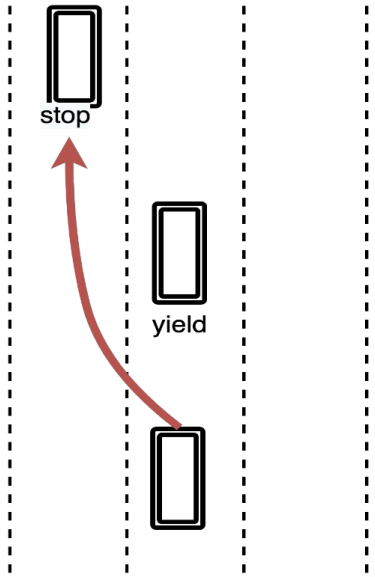
Path Planning and Optimization

PATH_DECIDER: make decisions based on the chosen lane (stop or yield)

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